

You may find the following formulae and values useful:

$$\Delta E = Pt \quad E_k = \frac{1}{2} mv^2 \quad E_p = mg\Delta h \quad W = Fd$$

$$E_{\text{thermal}} = mc\Delta T \quad E_{\text{thermal}} = mL$$

$$P = VI \quad V = IR$$

The specific heat capacity of water is $4,200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

The specific heat capacity of ice is $2,090 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

Latent heat of vapourisation for water is $2,300,000 \text{ J kg}^{-1}$

Latent heat of fusion for water is $334,000 \text{ J kg}^{-1}$

NAME & SURNAME: _____

2026 Topic Test AS92047

Physics, Earth & Space Science 1.4

(Adapted from CSTA 2025) Energy in a Physical System

QUESTION ONE

Ella is participating in the Tour Te Waipounamu cycle race. This is a multi-day, off road cycling race from the top of the South Island to the bottom.



<https://www.groundeffect.co.nz/tokos/dush-fund/18881527-hogs-back-track>

During part of the race, Ella cycles up the *Hog's Back* hill. This section has a climb of 150m. Ella and her bike have a combined mass of 76kg.

- Describe the energy changes that occur as Ella cycles up the hill.
- Calculate the change in gravitational potential energy that occurs.

As Ella begins cycling up the Hog's Back Hill, she comes to a junction in the track. One track is less steep, but longer and will take her 12 minutes and 30 seconds to get to the top. The other track is shorter and steeper, this track will take her 10 minutes and 20 seconds to climb to the top.

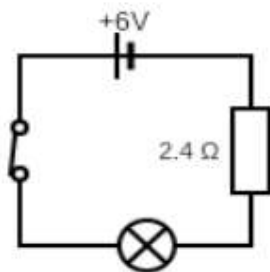
- Discuss which track will feel 'easier' for Ella to pedal her bike up.
Use calculations to support your answer.

Further on during the race, Ella approaches another small hill, 3.5m high. She is feeling tired and tries to 'coast' up the hill without pedalling.

- Calculate the speed that Ella must have theoretically been travelling for her to make it up the hill without pedalling.
Start by calculating the gravitational potential energy gained if she got to the top of the small hill.
- Ella is surprised that despite travelling at the speed calculated above, she does not make it to the top of the hill. Explain why this is the case.

QUESTION TWO

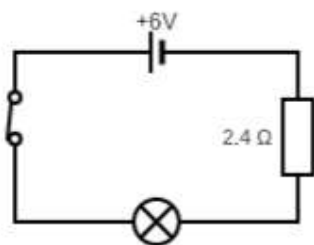
The Tour Te Waipounamu cycle race that Ella is participating in does not stop during the nighttime. To be able to ride in the dark, Ella has a powerful light mounted on the handlebars of her bike. The circuit for the light is shown below.



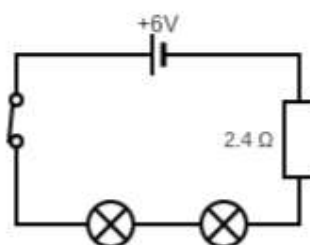
- Describe where the energy to power the light comes from.
- The circuit has a total resistance of $5.2\ \Omega$. Calculate the current through the lamp.
- Calculate the power output of the lamp.

Ella decides she needs her light to be brighter, so she adds another identical lamp to the circuit, as shown below in circuit two.

Circuit One:



Circuit Two:



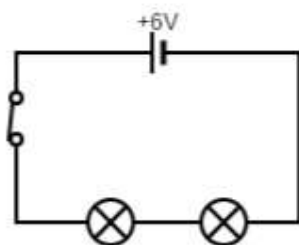
Ella is concerned that by adding the extra lamp, it might drain the cell faster. The cell in her original light, circuit one, would last approximately 3 hours.

The 6V cell has 75 000 J of energy stored in it. Each lamp has a resistance of $2.8\ \Omega$.

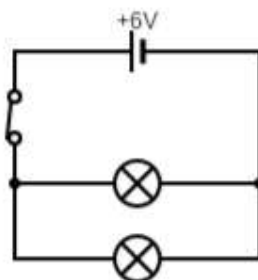
- Determine whether adding an extra lamp as shown in circuit two will drain the cell faster or slower. Support your answer by calculating the time it would take for circuit two to drain the cell.

Ella decides to connect the extra lamp to her circuit but removes the resistor. She has two different ways that she could connect it, shown below as either circuit three or circuit four. Each lamp has a resistance of $2.8\ \Omega$.

Circuit Three:



Circuit Four:



- e) Which way should Ella connect the extra lamp to the circuit to make the circuit brighter?
(Circle the correct answer below)

Circuit Three

Circuit Four

Explain why Ella should connect the lamps this way to make them as bright as possible. No calculations are necessary.

QUESTION THREE

During the cycle race, Ella has a drink bottle of water.

The drink bottle can hold 0.75kg of water.

Prior to starting the race, Ella places an ice cube in her drink bottle of water. The ice cube has a mass of 0.014kg. During the race, the ice cube slowly melts.



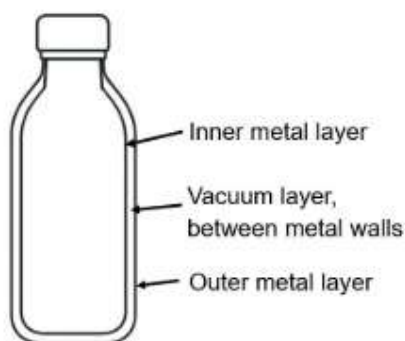
- a) Calculate the amount of energy required to melt the ice cube.
- b) Describe which direction the energy moves when the ice cube melts in the water.

For the second day of the race, Ella dissolves some salt into the water in her drink bottle to make an electrolyte drink. She then places it in the fridge to cool it down,

Ella's new electrolyte drink has a specific heat capacity of $4\,000\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- c) Explain whether Ella's new electrolyte drink would stay cooler for longer than pure water. Calculations are not required.

Ella's drink bottle is designed to keep liquids cool for as long as possible. The drink bottle is made of two layers of shiny metal. Between the two layers is a vacuum, where the air has been removed.



- d) Explain how the design features of Ella's drink bottle help to keep the contents cool for a long period of time. Include in your answer an explanation of how the cool water inside the drink bottle would slowly warm up and how the features of the water bottle help to prevent this.

For the third day of the race, Ella decides to completely freeze the water in her water bottle. She fills it up with 0.75kg of pure water and places it in the freezer. Before placing it in the freezer, the water in Ella's full drink bottle was 20°C .

- e) Calculate the amount of energy that would need to be removed from the water in the bottle to **completely** freeze all the water in the bottle.